

*Dr. R. Wampfler**C. Yang, X. Zhang, J. Jiang, Y. Zhang***Final Exam**

August 13th 2025

First and Last name: _____

ETH number: _____

Signature: _____

General Remarks

- Please check that your exam questionnaire is complete (there are 22 pages total).
- Remove all material from your desk which is not allowed by examination regulations.
- Fill in your first and last name and your ETH number and sign the exam. Place your student ID in front of you.
- You have 2 hours for the exam. There are 9 questions, where you can earn a total of 120 points. You don't have to score all points to obtain a very good grade.
- You can write your answers directly below the questions. Put your name and ETH number on top of each extra sheet that you use.
- Please do not use a pencil or red color pen to write your answers.
- You may provide at most one valid answer per question. Invalid solutions must be canceled out clearly.

	Topic	Max. Points	Points Achieved	Visum
1	Multiple Choice	16		
2	Affective Computing	13		
3	Deep Learning	13		
4	Speech Recognition	13		
5	Large Language Models	13		
6	Speech Synthesis	13		
7	Knowledge Graphs	13		
8	Reinforcement Learning	13		
9	Animation	13		
Total		120		

Question 1: Multiple Choice (16 pts.)

Mark the correct answer for the multiple-choice questions below. One point is given for the correct answer, one point is deducted for an incorrect answer. No point is given or deducted if a question has not been answered. The minimum number of points for the entire block of multiple-choice questions is 0.

- a) Artifact detection is essential in affective computing because physiological signals like EEG and ECG can be contaminated by noise from muscle movements or environmental interference. **1 pt.**
- ☐ True
☐ False
- b) In backpropagation, the gradient of the loss with respect to each weight is computed independently of the activation functions used in the network. **1 pt.**
- ☐ True
☐ False
- c) Discrete Fourier Transform (DFT) is commonly applied to short-time frames of a speech signal to analyze how frequency content changes over time. **1 pt.**
- ☐ True
☐ False
- d) Prosody refers exclusively to the spectral features of speech, such as formants and harmonics, without involving rhythm or intonation **1 pt.**
- ☐ True
☐ False
- e) Parameter-Efficient Fine-Tuning methods (PEFT) like LoRA and adapters reduce computational cost by freezing the majority of a pre-trained model's weights and learning only a small number of additional parameters. **1 pt.**
- ☐ True
☐ False
- f) Prompt engineering can be used to implicitly control the chatbot's tone, style, and persona without modifying the underlying model. **1 pt.**
- ☐ True
☐ False

g) Hybrid fusion strategies eliminate the need for modality-specific embedding by treating all modalities identically throughout the model pipeline. **1 pt.**

- ☐ True
- ☐ False

h) Stress depends on sentence context, while accent is fixed within the word. **1 pt.**

- ☐ True
- ☐ False

i) WaveNet uses causal, dilated convolutions to model long-range temporal dependencies in sequential data such as speech. **1 pt.**

- ☐ True
- ☐ False

j) In Linear Blend Skinning, each vertex is influenced by exactly one bone. **1 pt.**

- ☐ True
- ☐ False

k) Linear interpolation between animation key frames often results in unnatural or abrupt motion rather than smooth transitions. **1 pt.**

- ☐ True
- ☐ False

l) In normalizing flows, the model learns only the prior distribution p_Z , while the transformation functions remain fixed. **1 pt.**

- ☐ True
- ☐ False

m) An autonomous agent's actions are determined solely by its immediate perception of the environment. **1 pt.**

- ☐ True
- ☐ False

n) A primary disadvantage of dialogue trees is their limited flexibility, which makes it difficult for them to handle unexpected user queries or engage in open-ended conversations. **1 pt.**

- ☐ True
- ☐ False

o) In on-policy reinforcement learning, data collected from older versions of the policy can be stored in a replay buffer and reused to improve training efficiency. **1 pt.**

- ☐ True
- ☐ False

p) In the Q-learning algorithm, the policy is implicitly derived by selecting the action with the highest estimated Q-value for a given state, rather than being represented by a separate, explicit policy network **1 pt.**

- ☐ True
- ☐ False

Question 2: Affective Computing (13 pts.)

- a) Fig. 1 shows two different emotional responses to the same stimulus — the sight of an oncoming car. These cases represent two of the following emotion theories — James-Lange, Cannon-Bard, and Schachter-Singer (note that one theory is not depicted). For each case in the figure, identify the corresponding theory and briefly explain your reasoning. **4 pts.**

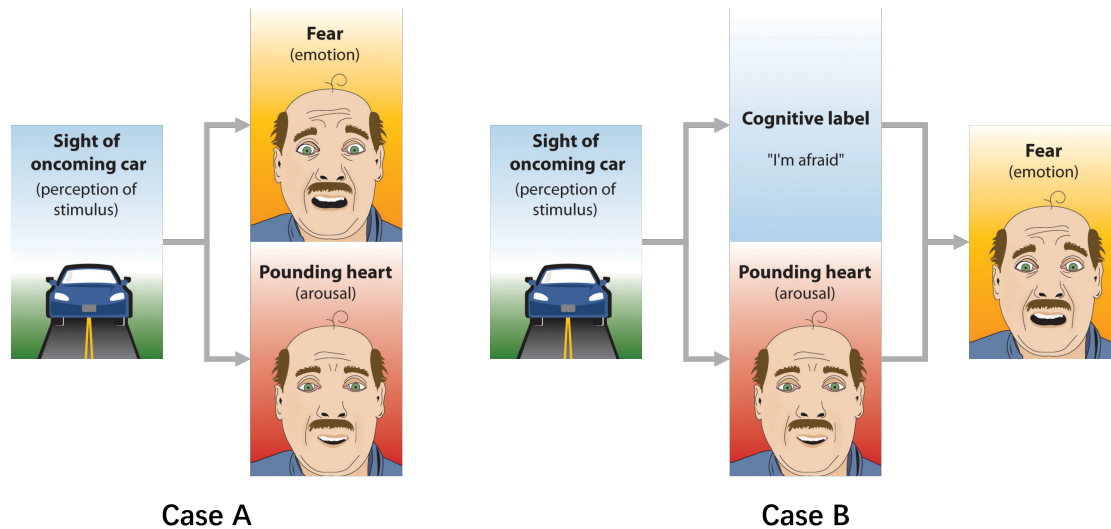


Figure 1: Examples of emotion theories.

- b) Imagine the emotional state "Anxious, tense, and helpless", how would you describe this emotional state using the dimensional model of emotion? Please describe with valence (positive/neutral/negative), arousal (high/moderate/low), and dominance (high/moderate/low), along with brief reasoning. **3 pts.**

c) Artifact detection is often applied to remove noise from physiological signals to ensure accurate analysis. Given a sequence of R-R intervals in milliseconds: {795, 800, 400, 810, 800, 805}, please answer the following questions:

i) What is the heart rate (beats per minute) for the R-R interval of 400 ms? **1 pt.**

ii) Assuming the last valid R-R interval equals 800 ms, please determine if the **400 ms R-R interval** constitutes an artifact based on the algorithm shown in the lecture. The Criterion Beat Difference (CBD) in the artifact detection algorithm is defined as

$$\begin{aligned} \text{CBD} &= \frac{\text{MAD} + \text{MED}}{2}, \quad \text{where} \\ \text{MAD} &= \frac{\text{Median Beat} - 2.9 \times \text{QD}}{3}, \\ \text{MED} &= 3.32 \times \text{QD}. \end{aligned}$$

The Quartile Deviation (QD) is defined as $\text{QD} = \frac{Q_3 - Q_1}{2}$, where Q_3 and Q_1 denote the third and the first quartile, respectively. You may assume that the lower and upper thresholds for valid R-R intervals are 300 ms and 2000 ms, respectively. Write down all **intermediate steps**. Round numbers to two decimal places. **5 pts.**

Question 3: Deep Learning (13 pts.)

- a) Recurrent Neural Networks (RNNs) are often used to process sequential or time-dependent data, such as texts and videos. These architectures are fundamental for character dialogue systems and speech synthesis. Please answer the following questions about RNNs:

i) Explain why RNNs are suitable for handling sequential data. **1 pt.**

ii) Explain why standard RNNs often suffer from the vanishing gradient problem. **1 pt.**

iii) Briefly describe the gating mechanism of Long Short-Term Memory (LSTM) modules, and explain how they mitigate the vanishing gradient problem. **2 pts.**

- b) Given a Transformer model processing a sequence of two tokens x_0 and x_1 , each with a two-dimensional embedding. The token embeddings are:

$$X = \begin{bmatrix} x_0 \\ x_1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

i) Use the sinusoidal positional encoding with:

$$PE_{(pos,0)} = \sin(pos), \quad PE_{(pos,1)} = \cos(pos)$$

Compute the positional encodings for positions 0 and 1, and add them to the input embeddings to get the final inputs to the attention layer (denoted as $\hat{X}$). Write down all **intermediate steps**. Round numbers to two decimal places. **3 pts.**

ii) Let the query, key, and value equal to $\hat{X}$, i.e.,

$$Q = K = V = \hat{X} \in \mathbb{R}^{2 \times 2}$$

Using the **scaled dot-product** attention formula, perform the following steps:

- Compute the attention scores (before softmax).
- Apply softmax to get attention weights.
- Briefly interpret the attention weights.
- Compute the output of the attention layer.

6 pts.

Question 4: Speech Recognition (13 pts.)

- a) In speech recognition systems, Word Error Rate (WER) is a common metric for evaluating transcription accuracy. Given are a reference transcription and the corresponding speech recognition output:

- **Reference:** the quick brown fox jumps over the lazy dog
- **Model Output:** the quick brown fax jump over lazy dogs

Please answer the following questions:

- Identify the number of substitutions, insertions, and deletions between the reference and the model output. **3 pts.**
- Compute the Word Error Rate (WER) for the above output and provide the formula used in your calculation. **2 pts.**
- What are the theoretical minimum and maximum possible values of WER? Explain briefly. **2 pts.**

- b) In speech recognition, the Discrete Fourier Transform (DFT) is a fundamental tool used to analyze the frequency content of speech signals. Consider a discrete-time signal of length 3, and its DFT results are as follows:

$$X[n] = [0, 1, 5], \quad \text{for } n = 0, 1, 2$$

- i) Compute the power spectrum P and explain the meaning of the power spectrum. Round numbers to two decimal places. **4 pts.**

- ii) Is the DFT typically applied to the entire speech signal or to short-time frames? Briefly explain why. **2 pts.**

Question 5: Large Language Models (13 pts.)

- a) Large Language Models (LLMs) can be adapted to specific tasks using various techniques. Two commonly used approaches are prompt engineering and fine-tuning. Briefly explain how each method works, and identify one difference in their mechanisms. **3 pts.**

- b) LLMs often use sampling strategies such as greedy decoding, top-k sampling, and top-p (nucleus) sampling to generate responses.

- i) Briefly explain the difference between top-k sampling and top-p (nucleus) sampling. **2 pts.**

- ii) Suppose a language model outputs the following probability distribution for the next token:

Token Index	Token	Probability
1	cat	0.35
2	dog	0.30
3	runs	0.15
4	fast	0.10
5	tree	0.06
6	blue	0.04

- Using top-p sampling with $p = 0.85$, please list the candidate tokens and provide all computation steps. **2 pts.**

c) Consider a language model generating two candidate sentences for a given prompt:

- Sentence A: The cat sits quietly.
- Sentence B: A dog lies under the table.

After applying the Word2Vec approach, the 3-dimensional sentence embedding vectors for sentence A (e_A) and B (e_B) are as follows:

$$e_A = [0.5, 0.1, 0.4], \quad e_B = [0.4, 0.3, 0.1].$$

The language model assigns the following probabilities to the tokens in Sentence A:

Previous Token	Next Token	Probability
<start>	The	0.20
The	cat	0.10
The cat	sits	0.15
The cat sits	quietly	0.25

- i) Compute the cosine similarity between e_A and e_B . Write down the **formula** and all **intermediate steps**. **4 pts.**

- ii) Compute the perplexity of Sentence A using the given token probabilities. **2 pts.**

Question 6: Speech Synthesis (13 pts.)

a) Classical Speech Synthesis Methods

i) Why is homograph disambiguation important in text normalization? Provide a concrete example and explain how a TTS system might resolve it. **2 pts.**

ii) Pronunciation dictionaries like CMU and Unisyn are commonly used in TTS systems. What is one limitation of relying solely on such dictionaries during phonetic analysis, and how can modern systems address this limitation? **2 pts.**

iii) Prosodic features such as pitch, duration, and prominence contribute to natural-sounding speech. Describe two types of prosodic phenomena. **2 pts.**

b) Acoustic Model, Vocoder, and Style Transfer

- i) Tacotron and FastSpeech differ in their generation strategies. Describe one key architectural difference between Tacotron and FastSpeech, and explain how this impacts both audio quality and inference speed. **2 pts.**

- ii) DiffWave uses diffusion models for high-quality speech synthesis. Explain the forward process, the reverse process, and identify what information is used to condition the generation process. **3 pts.**

- iii) You are designing a multi-speaker TTS system. Explain why you might prefer learned speaker embeddings over one-hot labels. Describe how such embeddings can be integrated into the acoustic model. **2 pts.**

Question 7: Knowledge Graphs (13 pts.)

- a) You are tasked with building a knowledge graph for a university based on the following paragraph.

Dr. Sarah Clark, a researcher at the Computer Science Department of ETH Zurich, recently published a paper titled "Advanced Machine Learning Algorithms" at the prestigious ICML conference. She is currently leading the "AI for Healthcare" research project, which is funded by her institution. Dr. Clark supervises graduate student Mark Thompson, who is also affiliated with the same department.

The knowledge graph has these 6 entity types: **Researcher**, **Publication**, **Project**, **Institution**, **Department**, **Conference**.

The knowledge graph schema is defined as following:

- (Researcher, *affiliatedWith*, Department)
- (Department, *partOf*, University)
- (Researcher, *published*, Publication)
- (Researcher, *supervise*, Researcher)
- (Publication, *presentedAt*, Conference)
- (Researcher, *lead*, Project)

The following portion of the knowledge graph has already been constructed based on the provided schema:

(Dr. Sarah Clark, *affiliatedWith*, Computer Science Department)

(Computer Science Department, *partOf*, ETH Zurich)

(Dr. Sarah Clark, *published*, "Advanced Machine Learning Algorithms")

("Advanced Machine Learning Algorithms", *presentedAt*, ICML)

(Dr. Sarah Clark, *lead*, "AI for Healthcare")

Add two more triples to complete the knowledge graph based on the provided schema. Then, identify two distinct types of relationships represented in the final knowledge graph.

4 pts.

b) You will now use the knowledge graph constructed in a) to answer the following query:
Query: "Find all researchers who are affiliated with the Computer Science Department AND have published papers that were presented at ICML conference AND are currently leading a research project."

i) Identify the type of query as discussed in the lecture. **1 pt.**

ii) Write a complete SPARQL query that retrieves the correct answer. Use the `uni` prefix for all relationships (e.g., `uni:affiliatedWith`). You may omit prefixes for entity names. **3 pts.**

c) You will apply the TransE embedding model to the knowledge graph from a). TransE represents entities and relations as vectors in a continuous space, where for a true triple (h, r, t) , the equation $\mathbf{h} + \mathbf{r} \approx \mathbf{t}$ should hold. Assume the following learned embeddings after training TransE on the knowledge graph:

- Dr. Sarah Clark = $[2.0, 1.5, -0.5]$
- Mark Thompson = $[1.8, 0.8, -0.2]$
- Computer Science Department = $[3.2, 2.1, 0.3]$
- ETH Zurich = $[4.0, 3.0, 1.0]$
- affiliatedWith = $[1.2, 0.6, 0.8]$
- supervise = $[-0.2, -0.7, 0.3]$
- partOf = $[0.8, 0.9, 0.7]$

i) Compute the score for the triple (Dr. Sarah Clark, supervise, Mark Thompson) using the TransE model. Specify the scoring function and show your steps. **2 pts.**

ii) Using the TransE model, evaluate whether the triple (Mark Thompson, affiliated-With, Computer Science Department) is likely to be true. Calculate the corresponding TransE score, compare it to the threshold of 2.0, and justify your conclusion based on this comparison. **1 pt.**

iii) Explain why the TransE model has difficulty representing symmetric relationships such as *supervise*, where both (Dr. Sarah Clark, supervise, Mark Thompson) and (Mark Thompson, supervise, Dr. Sarah Clark) are considered true facts. Use the TransE scoring function to provide a mathematical explanation, and describe the implication this has for the learned entity embeddings. **2 pts.**

Question 8: Reinforcement Learning (13 pts.)

We aim to train a humanoid robot to reach a target location using reinforcement learning (RL). The robot starts at position $\mathbf{x}_0 \in \mathbb{R}^2$ and must navigate to a target location $\mathbf{y}_g \in \mathbb{R}^2$ as quickly as possible. The robot's movement follows the dynamics $\mathbf{x}_t = f(\mathbf{x}_{t-1}, \mathbf{z}_t)$, where $\mathbf{z}_t$ denotes the control signal at time step t . Assume there are no obstacles between the robot and its goal.

- a) To solve this navigation task using RL, you must first formulate it as an RL problem. Define the following components: the state, the action, and the policy function. **4 pts.**

- b) To train the policy, we need to collect training data by running episodes. Starting from the initial position $\mathbf{x}_0$, we execute the policy for T time steps to collect a trajectory τ . For this goal-reaching task, define the reward function r_t (what the agent should be rewarded for) and the return $R(\tau)$ (how the overall success of a trajectory is evaluated). **2 pts.**

- c) We now shift to a dense reward setting where a reward is calculated at each time step. To evaluate a given state s , we consider executing N different actions. Following each action a_n for T time steps generates a trajectory τ_n . The reward at time step t for the n -th trajectory is denoted by $r_{t,n}$.

Based on this setup, please provide expressions for the following:

1. The state-action value function $Q(s, a_n)$, which represents the expected return after taking action a_n from state s .
2. The state value function $V(s)$. Assume the actions a_n are sampled from a known policy $\pi(a|s)$ and use the $Q(s, a_n)$ values you defined above.

4 pts.

- d) Explain three advantages of combining PPO with an Actor-Critic (e.g., A2C) architecture compared to the standard policy gradient method.

3 pts.

Question 9: Animation (13 pts.)

a) Traditional Animation Techniques

Your 2D character has an articulated body. Its structure consists of three bones that are connected through joints, where the root joint's location is fixed and located in $[0, 0]^T$. The length of the links are $L_1 = 2$, $L_2 = 2$, and $L_3 = 1$. All joints θ_1 , θ_2 , and θ_3 have one degree of freedom (DoF). Initially, at step $t = 0$, we have $\theta_1^0 = \frac{\pi}{6}$, $\theta_2^0 = \frac{\pi}{6}$, and $\theta_3^0 = \frac{\pi}{6}$. You want your character to reach a ball that is placed at $[5, 2]^T$, which means you need to bring the end-effector to $e^* = [5, 2]^T$ (see Figure 2). The following forward kinematics function can be assumed:

$$e^t = F(\theta_1^t, \theta_2^t, \theta_3^t) = \begin{bmatrix} L_1 \cdot \cos(\theta_1^t) + L_2 \cdot \cos(\theta_1^t + \theta_2^t) + L_3 \cdot \cos(\theta_1^t + \theta_2^t + \theta_3^t) \\ L_1 \cdot \sin(\theta_1^t) + L_2 \cdot \sin(\theta_1^t + \theta_2^t) + L_3 \cdot \sin(\theta_1^t + \theta_2^t + \theta_3^t) \end{bmatrix}$$

Perform one step of the Jacobian method used in inverse kinematics by filling in the missing steps in the following sub-tasks. You should use a learning rate of $\alpha = 0.1$. Write down all **formulas** and **intermediate steps**. You can round to two decimal places.

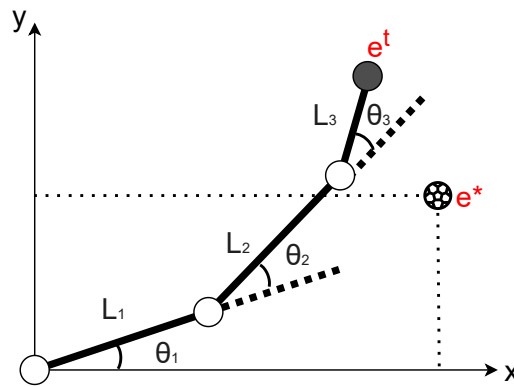


Figure 2: Illustration of the initial configuration of the character's arm.

- i) Compute the initial position of the end-effector, i.e.,

2 pts.

$$e^0 =$$

- ii) Compute the error in the end-effector position:

1 pt.

$$\Delta e =$$

iii) Calculate the Jacobian:

3 pts.

$$J =$$

iv) Compute the change in rotation angles. You can use the following Moore-Penrose inverse (pseudoinverse) of J :

2 pts.

$$J^+ = J^\top (JJ^\top)^{-1} \\ \approx \begin{bmatrix} -0.29 & 0.47 \\ -0.19 & 0.19 \\ -0.05 & 0.12 \end{bmatrix}.$$

$$\begin{bmatrix} \Delta\theta_1 \\ \Delta\theta_2 \\ \Delta\theta_3 \end{bmatrix} =$$

- v) Update the rotation angles (you do not need to compute the resulting new end-effector positions): **2 pts.**

$$\begin{bmatrix} \theta_1^1 \\ \theta_2^1 \\ \theta_3^1 \end{bmatrix} =$$

b) **Advanced animation techniques**

- i) Recent advances in data-driven animation have explored ways to model motion dynamics more efficiently. What is the core idea behind the DeepPhase approach? **1 pt.**
- ii) Explain the main idea of normalizing flows and name one application for character animation. **2 pts.**